

CYBERSECURITY PORTFOLIO

HANDS-ON LAB · IDENTITY & ACCESS SECURITY

Linux Identity & Access Hardening

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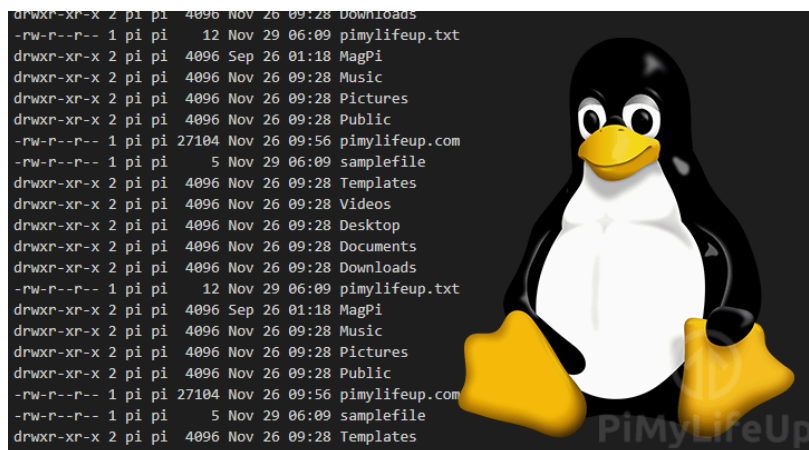
Portfolio Edition

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Securing User Accounts and File Permissions

Week 2, Day 3



Part 1: User Account Audit and Cleanup (User Account Lead)

1. List all local accounts (cat /etc/passwd) and identify any that are stale, duplicated, or no longer tied to an active team member.

```
student@LinuxTargetVM3:~$ pwd
/home/student
student@LinuxTargetVM3:~$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534:./nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:./usr/sbin/nologin
systemd-timesync:x:996:996:systemd Time Synchronization:./usr/sbin/nologin
dhcpcd:x:100:65534:DHCP Client Daemon,,./usr/lib/dhcpcd:/bin/false
messagebus:x:101:101:./nonexistent:/usr/sbin/nologin
syslog:x:102:102:./nonexistent:/usr/sbin/nologin
systemd-resolve:x:991:991:systemd Resolver:./usr/sbin/nologin
uuidd:x:103:103:./run/uuidd:/usr/sbin/nologin
tss:x:104:104:TPM software stack,,./var/lib/tpm:/bin/false
sshd:x:105:65534:./run/sshd:/usr/sbin/nologin
pollinate:x:106:1:./var/cache/pollinate:/bin/false
tcpdump:x:107:108:./nonexistent:/usr/sbin/nologin
landscape:x:108:109:./var/lib/landscape:/usr/sbin/nologin
fwupd-refresh:x:990:990:Firmware update daemon:/var/lib/fwupd:/usr/sbin/nologin
polkitd:x:989:989:User for polkitd:./usr/sbin/nologin
chrony:x:109:113:Chrony daemon,,./var/lib/chrony:/usr/sbin/nologin
student:x:1000:1000:Ubuntu:/home/student:/bin/bash
oldservice:x:1001:1001:./home/oldservice:/bin/sh
postfix:x:110:115:./var/spool/postfix:/usr/sbin/nologin
aide:x:988:988:Advanced Intrusion Detection Environment:/var/lib/aide:/usr/sbin/nologin
```

```
student@LinuxTargetVM3:~$ sudo lastlog -u oldservice
[sudo] password for student:
Username      Port      From
oldservice                                Latest
**Never logged in**
```

oldservice was identified as a potentially stale account. It has UID 1001, a valid login shell (/bin/sh), a home directory, and an active password, but lastlog shows that it has never logged in. The remaining entries are expected system/service accounts.

2. Check for accounts with UID 0 other than root, and for any accounts with no password set or with a blank/weak password.

```
student@LinuxTargetVM3:~$ pwd
/home/student
student@LinuxTargetVM3:~$ awk -F: '$3 == 0 {print $1, $3}' /etc/passwd
root 0
```

No authorized UID 0 accounts were found besides root.

```
student@LinuxTargetVM3:~$ pwd
/home/student
student@LinuxTargetVM3:~$ sudo awk -F: '($2 == "") {print $1}' /etc/shadow
```

No accounts with blank passwords were identified.

3. Remove or lock any account you determine shouldn't exist, documenting your reasoning for each.

```
student@LinuxTargetVM3:~$ pwd
/home/student
student@LinuxTargetVM3:~$ sudo passwd -l oldservice
passwd: password changed.
student@LinuxTargetVM3:~$ sudo passwd -S oldservice
oldservice L 2026-08-05 0 99999 7 -1
```

The oldservice account was locked because it appeared to be stale/orphaned. It had never logged in, had an active password, and was not tied to the active student user. Locking the account disables password-based access while preserving the account and its files for review.

4. Confirm every remaining account has an appropriately restricted home directory (750 or stricter).

```
student@LinuxTargetVM3:~$ pwd
/home/student
student@LinuxTargetVM3:~$ ls -ld /home/*
drwxr-x--- 2 oldservice oldservice 4096 Aug  5 17:14 /home/oldservice
drwxr-x--- 5 student    student    4096 Aug 11 18:55 /home/student
```

Both home directories located were configured with 750 permissions. This prevents users outside the owner and assigned group from accessing the directories.

Part 2: Password Policy and MFA (Authentication Lead)

1. Review and strengthen the existing password policy (minimum length, complexity, expiration) using PAM.

```
student@LinuxTargetVM3:~$ pwd
/home/student
student@LinuxTargetVM3:~$ grep -E "password" /etc/pam.d/common-password
grep: /etc/pam.d/common-password: No such file or directory
student@LinuxTargetVM3:~$ grep -E "password" /etc/pam.d/common-password
password            requisite                                pam_pwquality.so retry=3
password            [success=1 default=ignore]      pam_unix.so obscure use_authtok try_first_pass yescrypt
password            requisite                                pam_deny.so
password            required                                pam_permit.so
student@LinuxTargetVM3:~$ grep -E "password" /etc/pam.d/system-auth
grep: /etc/pam.d/system-auth: No such file or directory
student@LinuxTargetVM3:~$
```

```
student@LinuxTargetVM3:~$ cat /etc/security/pwquality.conf
# Configuration for systemwide password quality limits
# Defaults:
#
# Number of characters in the new password that must not be present in the
# old password.
# difok = 1
#
# Minimum acceptable size for the new password (plus one if
# credits are not disabled which is the default). (See pam_cracklib manual.)
# Cannot be set to lower value than 6.
# minlen = 12
#
# The maximum credit for having digits in the new password. If less than 0
# it is the minimum number of digits in the new password.
# dcredit = 0
#
# The maximum credit for having uppercase characters in the new password.
# If less than 0 it is the minimum number of uppercase characters in the new
# password.
# ucredit = 0
#
# The maximum credit for having lowercase characters in the new password.
# If less than 0 it is the minimum number of lowercase characters in the new
# password.
# lcredit = 0
#
# The maximum credit for having other characters in the new password.
# If less than 0 it is the minimum number of other characters in the new
# password.
# ocredit = 0
#
# The minimum number of required classes of characters for the new
# password (digits, uppercase, lowercase, others).
# minclass = 3
#
# The maximum number of allowed consecutive same characters in the new password.
# The check is disabled if the value is 0.
# maxrepeat = 0
#
# The maximum number of allowed consecutive characters of the same class in the
# new password.
# The check is disabled if the value is 0.
# maxclassrepeat = 0
```

First I checked the current complexity rules.

```
student@LinuxTargetVM3:~$ cat /etc/login.defs | grep -E "PASS_MAX_DAYS|PASS_MIN_DAYS|PASS_WARN_AGE"
PASS_MAX_DAYS    Maximum number of days a password may be used.
PASS_MIN_DAYS    Minimum number of days allowed between password changes.
PASS_WARN_AGE    Number of days warning given before a password expires.
PASS_MAX_DAYS    99999
PASS_MIN_DAYS     0
PASS_WARN_AGE     7
```

Then I checked the expiration defaults

```
student@LinuxTargetVM3:~$ chage -l student
Last password change                    : Aug 05, 2026
Password expires                        : never
Password inactive                       : never
Account expires                         : never
Minimum number of days between password change : 0
Maximum number of days between password change : 99999
Number of days of warning before password expires : 7
student@LinuxTargetVM3:~$
```

```
student@LinuxTargetVM3:~$ pwd
/home/student
student@LinuxTargetVM3:~$ sudo cp /etc/security/pwquality.conf /etc/security/pwquality.conf.bak
```

Before:

```
# Configuration for systemwide password quality limits
# Defaults:
#
# Number of characters in the new password that must not be present in the
# old password.
# difok = 1
#
# Minimum acceptable size for the new password (plus one if
# credits are not disabled which is the default). (See pam_cracklib manual.)
# Cannot be set to lower value than 6.
minlen = 12
#
# The maximum credit for having digits in the new password. If less than 0
# it is the minimum number of digits in the new password.
# dcredit = 0
#
# The maximum credit for having uppercase characters in the new password.
# If less than 0 it is the minimum number of uppercase characters in the new
# password.
# ucredit = 0
#
# The maximum credit for having lowercase characters in the new password.
# If less than 0 it is the minimum number of lowercase characters in the new
# password.
# lcredit = 0
#
# The maximum credit for having other characters in the new password.
# If less than 0 it is the minimum number of other characters in the new
# password.
# ocredit = 0
#
# The minimum number of required classes of characters for the new
# password (digits, uppercase, lowercase, others).
minclass = 3
#
# The maximum number of allowed consecutive same characters in the new password.
# The check is disabled if the value is 0.
# maxrepeat = 0
#
# The maximum number of allowed consecutive characters of the same class in the
# new password.
# The check is disabled if the value is 0.
# maxclassrepeat = 0
#
# Whether to check for the words from the passwd entry GECOS string of the user.
# The check is enabled if the value is not 0.
# gecheck = 0
```

After:

```
# Configuration for systemwide password quality limits
# Defaults:
#
# Number of characters in the new password that must not be present in the
# old password.
difok = 1
#
# Minimum acceptable size for the new password (plus one if
# credits are not disabled which is the default). (See pam_cracklib manual.)
# Cannot be set to lower value than 6.
minlen = 12
#
# The maximum credit for having digits in the new password. If less than 0
# it is the minimum number of digits in the new password.
dcredit = -1
#
# The maximum credit for having uppercase characters in the new password.
# If less than 0 it is the minimum number of uppercase characters in the new
# password.
ucredit = -1
#
# The maximum credit for having lowercase characters in the new password.
# If less than 0 it is the minimum number of lowercase characters in the new
# password.
lcredit = -1
#
# The maximum credit for having other characters in the new password.
# If less than 0 it is the minimum number of other characters in the new
# password.
ocredit = -1
#
# The minimum number of required classes of characters for the new
# password (digits, uppercase, lowercase, others).
minclass = 3
#
# The maximum number of allowed consecutive same characters in the new password.
# The check is disabled if the value is 0.
maxrepeat = 3
#
# The maximum number of allowed consecutive characters of the same class in the
# new password.
# The check is disabled if the value is 0.
maxclassrepeat = 0
#
```

```

student@LinuxTargetVM3:~$ cat /etc/security/pwquality.conf
# Configuration for systemwide password quality limits
# Defaults:
#
# Number of characters in the new password that must not be present in the
# old password.
difok = 1
#
# Minimum acceptable size for the new password (plus one if
# credits are not disabled which is the default). (See pam_cracklib manual.)
# Cannot be set to lower value than 6.
minlen = 12
#
# The maximum credit for having digits in the new password. If less than 0
# it is the minimum number of digits in the new password.
dcredit = -1
#
# The maximum credit for having uppercase characters in the new password.
# If less than 0 it is the minimum number of uppercase characters in the new
# password.
uccredit = -1
#
# The maximum credit for having lowercase characters in the new password.
# If less than 0 it is the minimum number of lowercase characters in the new
# password.
lccredit = -1
#
# The maximum credit for having other characters in the new password.
# If less than 0 it is the minimum number of other characters in the new
# password.
occredit = -1
#
# The minimum number of required classes of characters for the new
# password (digits, uppercase, lowercase, others).
minclass = 3
#
# The maximum number of allowed consecutive same characters in the new password.
# The check is disabled if the value is 0.
maxrepeat = 3
#
# The maximum number of allowed consecutive characters of the same class in the
# new password.
# The check is disabled if the value is 0.
maxclassrepeat = 0
#
# Whether to check for the words from the passwd entry GECOS string of the user.
# The check is enabled if the value is not 0.
gecoscheck = 0

```

Proof of change

```

student@LinuxTargetVM3:~$ sudo cp /etc/login.defs /etc/login.defs.bak
student@LinuxTargetVM3:~$ sudo nano /etc/login.defs
student@LinuxTargetVM3:~$ cat /etc/login.defs

```

Before:

```

#
# Password aging controls:
#
#      PASS_MAX_DAYS   Maximum number of days a password may be used.
#      PASS_MIN_DAYS   Minimum number of days allowed between password changes.
#      PASS_WARN_AGE   Number of days warning given before a password expires.
#
PASS_MAX_DAYS   99999
PASS_MIN_DAYS    0
PASS_WARN_AGE    7

```

After:

```

#
# Password aging controls:
#
#      PASS_MAX_DAYS   Maximum number of days a password may be used.
#      PASS_MIN_DAYS   Minimum number of days allowed between password changes.
#      PASS_WARN_AGE   Number of days warning given before a password expires.
#
PASS_MAX_DAYS    90
PASS_MIN_DAYS     7
PASS_WARN_AGE    14

```

This is what the change looked like after

2. Install and configure the Google Authenticator PAM module for SSH login, enrolling at least one test account.

```
student@LinuxTargetVM3:~$ sudo apt update
Hit:1 http://azure.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://azure.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Hit:3 http://azure.archive.ubuntu.com/ubuntu noble-backports InRelease
Get:4 http://azure.archive.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:5 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1191 kB]
Get:6 http://azure.archive.ubuntu.com/ubuntu noble-updates/main Translation-en [282 kB]
Get:7 http://azure.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1683 kB]
Get:8 http://azure.archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [335 kB]
Get:9 http://azure.archive.ubuntu.com/ubuntu noble-security/main amd64 Packages [930 kB]
Get:10 http://azure.archive.ubuntu.com/ubuntu noble-security/main Translation-en [201 kB]
Get:11 http://azure.archive.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1200 kB]
Get:12 http://azure.archive.ubuntu.com/ubuntu noble-security/universe Translation-en [239 kB]
Fetched 6314 kB in 2s (3449 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
```

```
student@LinuxTargetVM3:~$ sudo apt install libpam-google-authenticator
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
libpam-google-authenticator is already the newest version (20191231-2build1).
The following package was automatically installed and is no longer required:
  tcpd
Use 'sudo apt autoremove' to remove it.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
```



```
Your new secret key is: DKCTICWMU3HVFUXYHNOGBX5
Enter code from app (-1 to skip): -1
Code confirmation skipped
Your emergency scratch codes are:
34976983
97409604
35567302
73156080
53916539
```

```
auth required pam_google_authenticator.so
```

3. Update `sshd_config` to require both a key/password and the one-time code (ChallengeResponseAuthentication / AuthenticationMethods), then restart SSH.

```
#PubkeyAuthentication yes
Password Authentication yes
ChallengeResponseAuthentication yes|
# Expect .ssh/authorized_keys2 to be disregarded by default in futur
#AuthorizedKeysFile .ssh/authorized_keys .ssh/authorized_keys2
```

Part 3: File Permissions and Ownership Audit (File Permissions Lead)

1. Search for SUID and SGID binaries on the host: `find / -perm -4000 -o -perm -2000 -type f 2>/dev/null`, and determine whether each one is legitimately required.

```
student@LinuxTargetVM3:~$ find / -perm -4000 -o -perm -2000 -type f 2>/dev/null
/usr/bin/chage
/usr/bin/sudo
/usr/bin/mount
/usr/bin/gpasswd
/usr/bin/expiry
/usr/bin/passwd
/usr/bin/chsh
/usr/bin/crontab
/usr/bin/umount
/usr/bin/ssh-agent
/usr/bin/newgrp
/usr/bin/fusermount3
/usr/bin/dotlockfile
/usr/bin/chfn
/usr/bin/su
/usr/sbin/postdrop
/usr/sbin/postqueue
/usr/sbin/mount.cifs
/usr/sbin/unix_chkpwd
/usr/sbin/pam_extrausers_chkpwd
/usr/lib/dbus-1.0/dbus-daemon-launch-helper
/usr/lib/x86_64-linux-gnu/utempter/utempter
/usr/lib/polkit-1/polkit-agent-helper-1
/usr/lib/openssh/ssh-keysign
```

All of these directories and binaries are required

2. Search for world-writable files and directories, and correct any that shouldn't be writable by everyone.

```
root@LinuxTargetVM3:/home/student# sudo find / -xdev -type f -perm -o=w ! -path "/tmp/*" ! -path "/var/tmp/*" 2>/dev/null
```

```

/proc/15158/task/15158/attr/smack/current
/proc/15158/task/15158/attr/apparmor/current
/proc/15158/task/15158/attr/apparmor/exec
/proc/15158/attr/current
/proc/15158/attr/exec
/proc/15158/attr/fscrate
/proc/15158/attr/keycreate
/proc/15158/attr/sockcreate
/proc/15158/attr/smack/current
/proc/15158/attr/apparmor/current
/proc/15158/attr/apparmor/exec
/proc/15158/timerslack_ns
/proc/15164/task/15164/attr/current
/proc/15164/task/15164/attr/exec
/proc/15164/task/15164/attr/fscrate
/proc/15164/task/15164/attr/keycreate
/proc/15164/task/15164/attr/sockcreate
/proc/15164/task/15164/attr/smack/current
/proc/15164/task/15164/attr/apparmor/current
/proc/15164/task/15164/attr/apparmor/exec
/proc/15164/attr/current
/proc/15164/attr/exec
/proc/15164/attr/fscrate
/proc/15164/attr/keycreate
/proc/15164/attr/sockcreate
/proc/15164/attr/smack/current
/proc/15164/attr/apparmor/current
/proc/15164/attr/apparmor/exec
/proc/15164/timerslack_ns

```

No world-writable files. Using the broader command uncovered over 200 files that were world writable. These were most likely virtual kernel files.

3. Audit ownership of key application and configuration directories, correcting any files owned by the wrong

user or group.

```

root@LinuxTargetVM3:/home/student# find /etc /usr/local /opt -xdev \( ! -user root -o ! -group root \) -ls 2>/dev/null
 1552      4 drwxrwxr-x  2 root    landscape  4096 Aug 21  2025 /etc/landscape
34334      4 -rw-r-----  1 root    shadow    1117 Aug 10 18:22 /etc/shadow-
34132      4 -rw-r-----  1 root    shadow    783 Aug 10 18:21 /etc/gshadow-
1585      4 drwxr-x---  2 root    polkitd   4096 Apr 10 10:57 /etc/polkit-1/rules.d
 902      4 drwxr-x---  2 root    ssl-cert  4096 Aug 10 18:21 /etc/ssl/private
34889      4 -rw-r-----  1 root    ssl-cert 1704 Aug 10 18:21 /etc/ssl/private/ssl-cert-snakeoil.key
76870      4 -rw-r-----  1 root    shadow    794 Aug 10 18:22 /etc/gshadow
  20      4 -rw-r-----  1 root    _chrony   481 Oct  2  2024 /etc/chrony/chrony.keys
 4359      4 -rw-r-----  1 root    shadow    1118 Aug 12 19:07 /etc/shadow
34543      0 -rw-----  1 oldservice oldservice 0 Aug  7 13:42 /etc/old_report.log

```

Oldservice has write access

4. Document every permission or ownership change with before-and-after values.

Before:

```

34543      0 -rw--w----  1 oldservice oldservice 0 Aug  7 13:42 /etc/old_report.log

```

After

```
root@LinuxTargetVM3:/etc# ls -l old_report.log  
-rw----- 1 root root 0 Aug  7 13:42 old_report.log
```

Ownership changed by using `chown root:root old_report.log` and `chmod 600 old_report.log`

Part 4: Access Control Lists for a Shared Directory (ACL Lead)

```
root@LinuxTargetVM3:~# pwd
/root
root@LinuxTargetVM3:~# sudo mkdir -p /srv/shared-cases
root@LinuxTargetVM3:~# sudo chmod 700 /srv/shared-cases
root@LinuxTargetVM3:~#
```

Create directory and lock permissions so unauthorized users cannot access.

```
root@LinuxTargetVM3:~# cd /srv/shared-cases
root@LinuxTargetVM3:/srv/shared-cases# sudo groupadd loanops
root@LinuxTargetVM3:/srv/shared-cases# sudo groupadd fraudinvestigations
root@LinuxTargetVM3:/srv/shared-cases#
```

Create **loanops** and **fraudinvestigations** groups.

```
root@LinuxTargetVM3:/srv/shared-cases# useradd -m -s /bin/bash loan_user1
root@LinuxTargetVM3:/srv/shared-cases# useradd -m -s /bin/bash fraud_user1
root@LinuxTargetVM3:/srv/shared-cases#
```

Create a test user account for each group.

```
root@LinuxTargetVM3:/srv/shared-cases# sudo passwd loan_user1
New password:
Retype new password:
passwd: password updated successfully
root@LinuxTargetVM3:/srv/shared-cases# sudo passwd fraud_user1
New password:
Retype new password:
passwd: password updated successfully
root@LinuxTargetVM3:/srv/shared-cases#
```

Set passwords for each user account.

- Trytohackme1!
- Trytohackme2@

```
root@LinuxTargetVM3:/srv/shared-cases# sudo usermod -aG loanops loan_user1
root@LinuxTargetVM3:/srv/shared-cases# sudo usermod -aG fraudinvestigations fraud_user1
root@LinuxTargetVM3:/srv/shared-cases#
```

Add user accounts to their respective groups.

```
root@LinuxTargetVM3:/srv/shared-cases# sudo setfacl -m g:loanops:rwx /srv/shared-cases
root@LinuxTargetVM3:/srv/shared-cases# sudo setfacl -m g:fraudinvestigations:rx /srv/shard-cases
setfacl: /srv/shard-cases: No such file or directory
root@LinuxTargetVM3:/srv/shared-cases# sudo setfacl -m g:fraudinvestigations:rx /srv/shared-cases
root@LinuxTargetVM3:/srv/shared-cases#
```

Grant **loanops** read, write, and execute permissions and give **fraudinvestigations** group read and execute.

```
root@LinuxTargetVM3:/srv/shared-cases# sudo setfacl -d -m g:loanops:rwx /srv/shared-cases
root@LinuxTargetVM3:/srv/shared-cases# sudo setfacl -d -m g:fraudinvestigations:rx /srv/shared-cases
root@LinuxTargetVM3:/srv/shared-cases#
```

Give default permissions to future files that are created in **loanops** and **fraudinvestigations**.

```
root@LinuxTargetVM3:/srv/shared-cases# getfacl /srv/shared-cases
getfacl: Removing leading '/' from absolute path names
# file: srv/shared-cases
# owner: root
# group: root
user::rwx
group::---
group:loanops:rwx
group:fraudinvestigations:r-x
mask::rwx
other::---
default:user::rwx
default:group::---
default:group:loanops:rwx
default:group:fraudinvestigations:r-x
default:mask::rwx
default:other::---
```

Confirm set permissions for created groups.

```
root@LinuxTargetVM3:~#  
logout  
student@LinuxTargetVM3:~$ su - loan_user1  
Password:  
loan_user1@LinuxTargetVM3:~$ cd /srv/shared-cases  
loan_user1@LinuxTargetVM3:/srv/shared-cases$ touch loan_test.txt  
loan_user1@LinuxTargetVM3:/srv/shared-cases$ ls -la  
total 12  
drwxrwx---+ 2 root      root      4096 Aug 12 19:48 .  
drwxr-xr-x  3 root      root      4096 Aug 12 19:27 ..  
-rw-rw----+ 1 loan_user1 loan_user1    0 Aug 12 19:48 loan_test.txt  
loan_user1@LinuxTargetVM3:/srv/shared-cases$ ls  
loan_test.txt  
loan_user1@LinuxTargetVM3:/srv/shared-cases$
```

Confirm permissions work for **loan_user1**.

```
loan_user1@LinuxTargetVM3:/srv/shared-cases$  
logout  
student@LinuxTargetVM3:~$ su - fraud_user1  
Password:  
fraud_user1@LinuxTargetVM3:~$ cd /srv/shared-cases  
fraud_user1@LinuxTargetVM3:/srv/shared-cases$ ls -la  
total 12  
drwxrwx---+ 2 root      root      4096 Aug 12 19:48 .  
drwxr-xr-x  3 root      root      4096 Aug 12 19:27 ..  
-rw-rw----+ 1 loan_user1 loan_user1    0 Aug 12 19:48 loan_test.txt  
fraud_user1@LinuxTargetVM3:/srv/shared-cases$ touch fraud_test.txt  
touch: cannot touch 'fraud_test.txt': Permission denied  
fraud_user1@LinuxTargetVM3:/srv/shared-cases$
```

Verify permissions work for **fraud_user1**.

Reflection Questions

1. What was the hardest part of dividing this work across the team, and how did you keep everyone's changes from conflicting on the shared host?

Dividing the work amongst ourselves was not so much the difficult part. To prevent changes from conflicting on the shared host we limited access to the virtual machine to only two group members at a time, which made getting the lab done in a timely manner a little more difficult.

2. Why do ACLs solve the shared-directory problem better than standard Linux owner/group/other permissions alone?

Standard Linux permissions allow only a single owner and group to be assigned to a file or directory. ACL's allow for fine grained role based access control where there are multiple groups and users that can be assigned specific/independent permissions on the same directory. ACL's also support default ACL's that automatically enforce inherited permissions on all newly created files.

3. How does adding MFA change the risk calculus for an account whose password is eventually guessed or leaked?

Adding MFA breaks the single point of failure model by having two factors that an attacker would have to go through. It overall changes the directory of the attack due to having to defeat another factor.